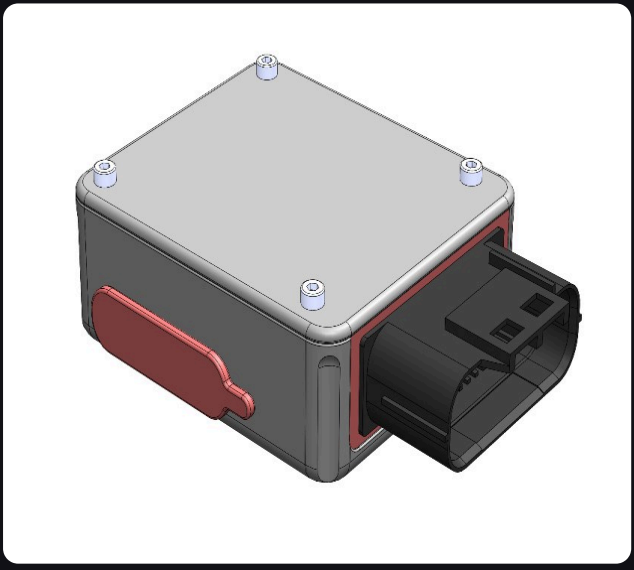
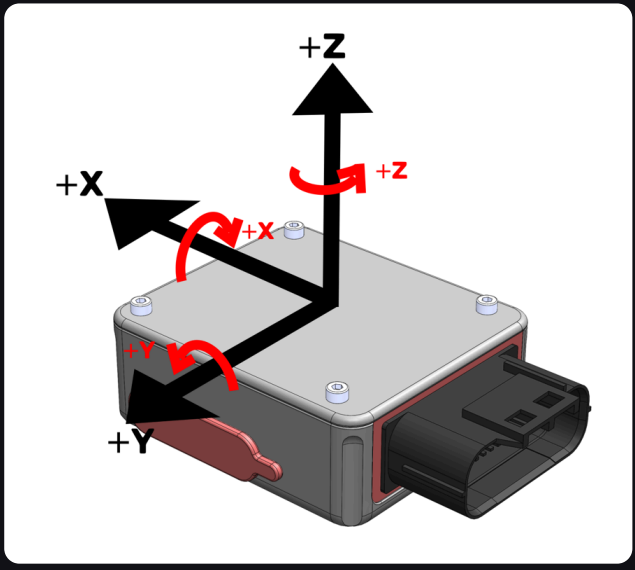
# GUBENAV

## Technical datasheet and manual

100 Hz GPS and 6-axis IMU sensor with CAN 2.0B interface · Revision 0014 · November 2026






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# 1 Overview

## 1.1 Description

**GUBENAV** is a vehicle position and dynamics sensor for those who need fast and reliable telemetry data. It combines a high-sensitivity **GPS/GNSS** receiver with an ICM-20649 6-axis inertial platform (**IMU**): with a *sensor fusion* algorithm it calculates and transmits geographic coordinates, speed, heading, accelerations and angular rates on the three axes with an update rate of up to **100 Hz**.

Data is transmitted in real time on the **CAN 2.0B** bus, with the proprietary GUBENAV protocol or with the **NMEA2000** standard: it is recorded with a **GUBELOG** or **MICROLOG** data logger or with any CAN acquisition system. Configuration is done with the free **GUBELLINI DataStudio** software via USB-C.

## 1.2 Main features

- **GPS/GNSS** with integrated antenna and **6-axis IMU** (accelerometers ±30 g, gyroscopes ±2000 °/s).
- **Sensor fusion** at 200 Hz; output selectable at **1, 10, 50 or 100 Hz**.
- Latitude, longitude, speed, heading, 3 accelerations and 3 angular rates.
- High-speed **CAN 2.0B** at 1 Mbps, 500 kbps or 250 kbps, 11- or 29-bit IDs; 4 programmable CAN IDs.
- **GUBENAV** (proprietary) or **NMEA2000** protocol (PGN 129025 and 129026).
- Sensor fusion filters optimised for car, motorcycle or boat.
- Power supply from **9 to 24 V DC**, 26-way **JAE MX23A** automotive connector, PA12 + 35% GF housing.

## 1.3 Applications

| Sector | Use |
|---|---|
| Motorsport and vehicle dynamics | High-speed trajectory, lean angles, accelerations for track analysis and vehicle development. |
| Precision agriculture | Tractor guidance systems, implement tracking and autonomous agricultural machinery with the NMEA2000 standard. |
| Industry and heavy machinery | Movement, speed and orientation of automated guided vehicles (AGVs), excavators and industrial robots. |
| Marine | Position, heading and speed on on-board NMEA2000 networks. |
| Laboratory and R&D | High-frequency position and dynamics, synchronised with other analog and digital sensors. |

# 2 Technical data

## 2.1 General specifications

| | Value |
|---|---|
| Positioning module | GPS/GNSS with integrated antenna |
| Inertial sensor | ICM-20649, 6 axes (3 accelerometers, 3 gyroscopes) |
| Sensor fusion | 200 Hz |
| Output rate | 1, 10, 50 or 100 Hz (selectable) |
| Output data | Latitude, longitude, speed, heading, accelerations X, Y, Z, angular rates X, Y, Z (roll, pitch, yaw), GPS fix validity |
| Interfaces | Data on CAN 2.0B; configuration via USB-C |

## 2.2 GPS/GNSS receiver

Integrated GPS/GNSS module with internal antenna. After power-on outdoors the first fix (cold start) can take up to 2 minutes; when there is no fix, GUBENAV keeps transmitting the IMU data and reports it in the fix validity field (§ 5.1).

## 2.3 Inertial platform (IMU)

6-axis sensor rigidly mounted on the internal circuit board: it measures longitudinal, lateral and vertical accelerations and roll, pitch and yaw rates.

| | Full scale | Internal sampling | Low-pass filter (digital IIR) |
|---|---|---|---|
| Accelerometers (3 axes) | ±30 g | 1125 Hz | 68.8 Hz |
| Gyroscopes (3 axes) | ±2000 °/s | 1125 Hz | 73.3 Hz |

## 2.4 CAN bus

| | Value |
|---|---|
| Transceiver | High-Speed CAN 2.0B |
| Bus speed | 1 Mbps, 500 kbps, 250 kbps |
| ID format | 11-bit (standard) or 29-bit (extended) |
| Protocols | GUBENAV (proprietary, 4 messages on 4 programmable IDs) or NMEA2000 (PGN 129025 and 129026, 29-bit IDs) |
| Termination | To be provided on the CAN line: a 120 Ω resistor at each end (GUBELOG and MICROLOG integrate it) |

## 2.5 Power supply and USB-C

| | Value |
|---|---|
| Supply voltage | From 9 to 24 V DC |
| Current consumption | 150 mA |
| USB-C | Configuration with GUBELLINI DataStudio (FTDI virtual serial port); during configuration GUBENAV is powered by the USB |

## 2.6 JAE MX23A26NF1 connector

Power and CAN pass through the 26-way **JAE Electronics MX23A26NF1** automotive connector, the same as on GUBELOG, watertight and with a vibration-resistant latch. Only 4 pins are used.

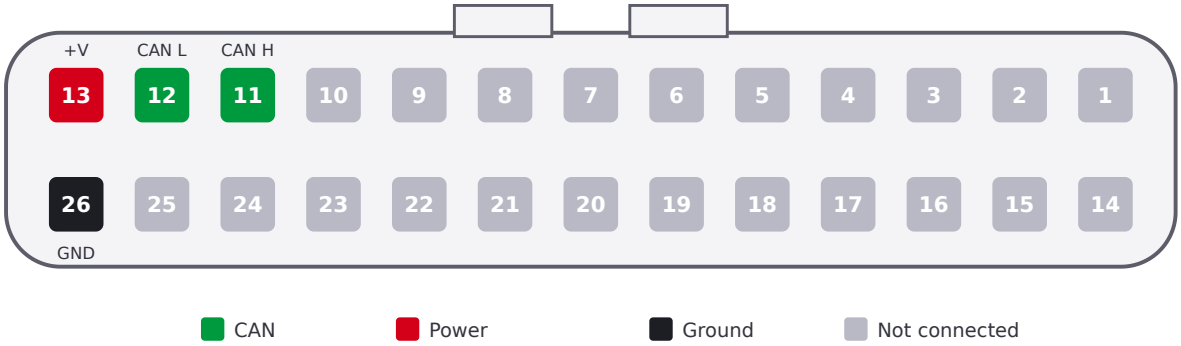



**Figure 1 –** GUBENAV connector pins, front view (as in the JAE manufacturer's drawing).

| Pin | Signal | Type |
|---|---|---|
| 11 | CAN H | CAN bus |
| 12 | CAN L | CAN bus |
| 13 | +V (9-24 V DC) | Power supply |
| 26 | GND | Ground |

> **Same connector as GUBELOG**
>
> The CAN H, CAN L, power and ground pins are the same as on GUBELOG: CAN H to CAN H, CAN L to CAN L, power and ground in common.

## 2.7 Operating environment

| | Value |
|---|---|
| Degree of protection | IP67 |
| Operating temperature | From -20 °C to +85 °C |

## 2.8 Dimensions and weight

| | **Value** |
|---|---|
| Housing | Nylon PA12 + 35% GF |
| Width × height | 63 × 27 mm (+2.5 mm for the USB cap) |
| Length (connector included) | 95.3 mm (body 70.5 mm) |
| Weight | On request |

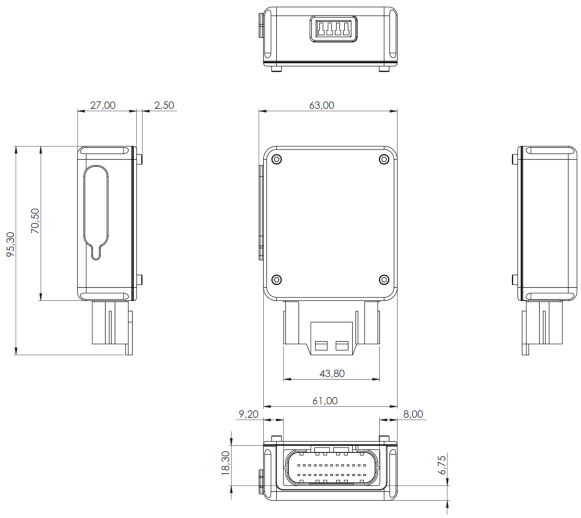



**Figure 2 –** GUBENAV dimensions (mm).

# 3 Installation

## 3.1 Position

- Mount GUBENAV where its top side has the clearest possible view of the sky (roof, fairing, roll bar).
- Avoid metal panels, thick carbon fibre and the vicinity of strong sources of electromagnetic interference (ignition coils, power inverters, transmitting antennas).

## 3.2 Orientation

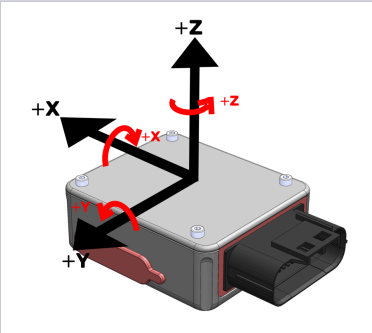



**Figure 3 –** GUBENAV measurement axes.

The accelerometers and gyroscopes measure along the sensor axes: GUBENAV must be mounted aligned and square with the vehicle, orthogonal to its axes.

- **X+** towards the direction of travel.
- The other axes as shown on the housing and in the figure.
- Positive rotations follow the red arrows (roll about X, pitch about Y, yaw about Z).

## 3.3 Vibration

Fix GUBENAV rigidly to the chassis or to a structural bracket not subject to flexing or resonance. Avoid soft plastics and moving fairings, which add mechanical noise to the accelerometer data. With very strong vibration (single-cylinder engines, heavy agricultural machinery) consider light anti-vibration mounts (silent blocks) for electronics.

# 4 Configuration with GUBELLINI DataStudio

## 4.1 The software

| Software | What it does |
|---|---|
| **GUBELLINI DataStudio** (free) | Configures Gubellini Electronics devices (GUBENAV, GUBELOG, MICROLOG, CAN boards); imports signals from DBC files; downloads datasets from the data loggers and exports to MoTeC, AEM and CSV; includes an AI assistant for device connection and configuration. |
| **GUBELLINI DataAnalysis** (free) | Analyses the recorded datasets: plots versus time or distance, layouts with multiple plots, math channels, track map, on-board video; it includes an **AI assistant** that analyses the data, writes reports and generates the code of math channels. |

## 4.2 Connection

1. Install DataStudio and connect GUBENAV to the PC with the USB-C cable: the USB powers the sensor and Windows installs the FTDI virtual serial port driver.
2. In the main window press **Sensor configuration** and choose **GUBENAV** in the list on the left.
3. At the bottom, under **Port**, choose the GUBENAV COM port (the button next to it refreshes the list).
4. **Read configuration** reads the parameters from the sensor; after making changes, **Send configuration** writes them to the GUBENAV permanent memory. The log at the bottom reports the result of each operation.

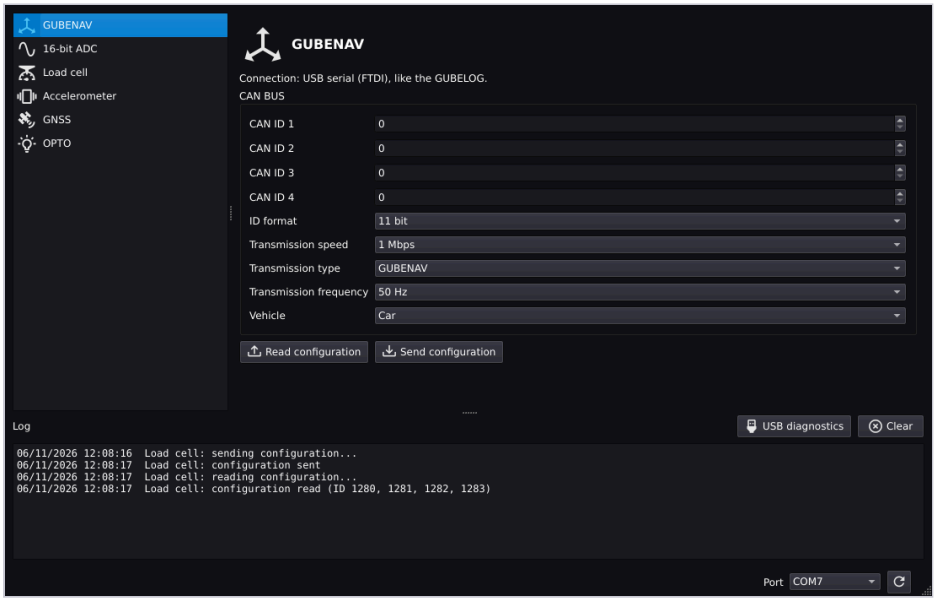



**Figure 4 –** The GUBENAV page in Sensor configuration.

## 4.3 Parameters

| Parameter | Meaning |
|---|---|
| **CAN ID 1** - **CAN ID 4** | The 4 IDs on which GUBENAV transmits the GUBENAV protocol messages, to be chosen so as to avoid conflicts with other ECUs on the network. In DataStudio they are entered in **decimal**: for example 256, 257, 258 and 259 correspond to 0x100-0x103. |
| **ID format** | 11-bit (standard) or 29-bit (extended), according to the receiver (e.g. GUBELOG, MICROLOG or third-party devices). |
| **Transmission speed** | 1 Mbps, 500 kbps or 250 kbps: all devices on the bus must use the same speed. |
| **Transmission type** | **GUBENAV** (proprietary protocol, § 5.1) or **NMEA2000** (§ 5.2). |
| **Transmission frequency** | 1, 10, 50 or 100 Hz, according to the load allowed on the bus. |
| **Vehicle** | Car, Motorbike or Boat: adapts the sensor fusion filters to the expected dynamics. |

> **Bus load**
> At 100 Hz the GUBENAV protocol sends 400 messages per second: at 250 kbps, with other devices on the same line, it is advisable to go down to 50 Hz.

# 5 CAN protocols

## 5.1 GUBENAV protocol

An efficient protocol, suitable for motorsport and general data acquisition. Each update is split into 4 messages of 8 bytes on the 4 configured IDs; data is in **Intel (little endian)** format.

### CAN ID 1 – GPS coordinates

| Byte | Data | Format | Conversion |
|---|---|---|---|
| 0-3 | Latitude | Signed 32-bit integer | value / 10,000,000 = degrees |
| 4-7 | Longitude | Signed 32-bit integer | value / 10,000,000 = degrees |

### CAN ID 2 – Kinematics and status

| Byte | Data | Format | Conversion |
|---|---|---|---|
| 0-1 | Speed | Unsigned 16-bit integer | value / 100 = km/h (resolution 0.01 km/h) |
| 2-3 | Heading | Unsigned 16-bit integer | value / 100 = degrees (resolution 0.01°) |
| 4 | Data validity | Byte | 1 = valid GPS fix, 0 = GPS timeout (IMU only) |
| 5-7 | Reserved | 3 bytes | 0 |

### CAN ID 3 – Accelerometers

| Byte | Data | Format | Conversion |
|---|---|---|---|
| 0-1 | Acceleration X | Signed 16-bit integer | value / 1000 = g |
| 2-3 | Acceleration Y | Signed 16-bit integer | value / 1000 = g |
| 4-5 | Acceleration Z | Signed 16-bit integer | value / 1000 = g |
| 6-7 | Reserved | 2 bytes | 0 |

### CAN ID 4 – Gyroscopes

| Byte | Data | Format | Conversion |
|---|---|---|---|
| 0-1 | Angular rate X | Signed 16-bit integer | value / 10 = °/s |
| 2-3 | Angular rate Y | Signed 16-bit integer | value / 10 = °/s |
| 4-5 | Angular rate Z | Signed 16-bit integer | value / 10 = °/s |
| 6-7 | Reserved | 2 bytes | 0 |

## 5.2 NMEA2000 protocol

In NMEA2000 mode GUBENAV transmits the standard messages (PGNs) used in the marine sector, in agricultural machinery and in fleets, with 29-bit IDs, priority 2 and fixed source address **0x33**.

## PGN 129025 – Position (ID 0x09F50333, 8 bytes)

| Byte | Data | Format | Notes |
|---|---|---|---|
| 0-3 | Latitude | Signed 32-bit integer | 1e-7 degrees, from the sensor fusion |
| 4-7 | Longitude | Signed 32-bit integer | 1e-7 degrees, from the sensor fusion |

## PGN 129026 – COG/SOG course and speed (ID 0x09F50433, 8 bytes)

| Byte | Data | Format | Notes |
|---|---|---|---|
| 0 | SID | Byte | 0 |
| 1 | COG reference | Byte | 0 = true north |
| 2-3 | COG (course over ground) | Unsigned 16-bit integer | resolution 0.0001 rad |
| 4-5 | SOG (speed over ground) | Unsigned 16-bit integer | resolution 0.01 m/s |
| 6-7 | Reserved | 2 bytes | 0xFFFF |

# 6 Recording with GUBELOG and MICROLOG

With the GUBENAV protocol the sensor data is recorded as CAN channels of a GUBELOG or MICROLOG. Connect CAN H, CAN L and ground of GUBENAV to the logger's CAN line and set in the logger the same speed and the same ID format as GUBENAV.

## 6.1 DBC file import

The quickest way is the `gubenav.dbc` file distributed with this datasheet (its text is in § 6.3), written for the default IDs 256-259 (0x100-0x103):

1. In the logger window (**GUBELOG** or **MICROLOG** button) choose **File → Import DBC…** and open `gubenav.dbc`.
2. Select the signals to record (11 in all) and assign them to channels CAN1-CAN16; DataStudio fills in CAN ID, bits, format, multiplier, label and unit.
3. If different IDs are used in GUBENAV, correct the CAN IDs in the logger table; then send the configuration to the logger.

## 6.2 CAN channels set manually

Alternatively, set the CUSTOM channels as follows (Intel / little endian format, offset 0):

| Channel | Label | CAN ID | Start bit | Bits | Type | Multiplier | Unit |
|---|---|---|---|---|---|---|---|
| CAN1 | Latitude | 256 | 0 | 32 | Signed | 0.0000001 | deg |
| CAN2 | Longitude | 256 | 32 | 32 | Signed | 0.0000001 | deg |
| CAN3 | Speed | 257 | 0 | 16 | Unsigned | 0.01 | km/h |
| CAN4 | Heading | 257 | 16 | 16 | Unsigned | 0.01 | deg |
| CAN5 | GPS_Valid | 257 | 32 | 8 | Unsigned | 1 | |
| CAN6 | Acc_X | 258 | 0 | 16 | Signed | 0.001 | g |
| CAN7 | Acc_Y | 258 | 16 | 16 | Signed | 0.001 | g |
| CAN8 | Acc_Z | 258 | 32 | 16 | Signed | 0.001 | g |
| CAN9 | Gyro_X | 259 | 0 | 16 | Signed | 0.1 | deg/s |
| CAN10 | Gyro_Y | 259 | 16 | 16 | Signed | 0.1 | deg/s |
| CAN11 | Gyro_Z | 259 | 32 | 16 | Signed | 0.1 | deg/s |

**Frequencies**

Set the logger channels to the same frequency as GUBENAV (for example 50 Hz): a higher frequency adds no information and takes up more space on the microSD.

## 6.3 gubenav.dbc file

```
VERSION ""

NS_ :
```

```
BS_:

BU_: GUBENAV Logger

BO_ 256 GUBENAV_Position: 8 GUBENAV
 SG_ Latitude : 0|32@1- (0.0000001,0) [-90|90] "deg" Logger
 SG_ Longitude : 32|32@1- (0.0000001,0) [-180|180] "deg" Logger

BO_ 257 GUBENAV_Kinematics: 8 GUBENAV
 SG_ Speed : 0|16@1+ (0.01,0) [0|655.35] "km/h" Logger
 SG_ Heading : 16|16@1+ (0.01,0) [0|360] "deg" Logger
 SG_ GPS_Valid : 32|8@1+ (1,0) [0|1] "" Logger

BO_ 258 GUBENAV_Accelerometer: 8 GUBENAV
 SG_ Acc_X : 0|16@1- (0.001,0) [-30|30] "g" Logger
 SG_ Acc_Y : 16|16@1- (0.001,0) [-30|30] "g" Logger
 SG_ Acc_Z : 32|16@1- (0.001,0) [-30|30] "g" Logger

BO_ 259 GUBENAV_Gyroscope: 8 GUBENAV
 SG_ Gyro_X : 0|16@1- (0.1,0) [-2000|2000] "deg/s" Logger
 SG_ Gyro_Y : 16|16@1- (0.1,0) [-2000|2000] "deg/s" Logger
 SG_ Gyro_Z : 32|16@1- (0.1,0) [-2000|2000] "deg/s" Logger

CM_ BU_ GUBENAV "GUBENAV GPS 100 Hz and 6-axis IMU sensor, GUBENAV protocol (default CAN IDs 0x100-0x103)";
CM_ SG_ 257 GPS_Valid "1 = valid GPS fix, 0 = GPS timeout (IMU only)";
VAL_ 257 GPS_Valid 1 "FIX_OK" 0 "IMU_ONLY" ;
```

# 7 Troubleshooting

| Problem | Cause and solution |
|---|---|
| No data on the CAN network | Check the 120 Ω termination resistors at the ends of the CAN line and that the speed set in DataStudio is the same as the receiver's; check ID and ID format in the logger. |
| GUBENAV does not appear in Sensor configuration | Use a USB-C data cable, check the FTDI driver and press the port refresh button. |
| Invalid or slow GPS coordinates | The sensor must not be covered by shielding materials; outdoors, wait for the first fix (up to 2 minutes). The GPS_Valid channel reads 0 when there is no fix. |
| Noisy accelerations | Check how rigidly the sensor is fixed: an unsuitable mount amplifies engine vibration and saturates the IMU. |
| Inverted signs or axes | Check the sensor orientation (§ 3.2). |

# 8 Ordering information

| Code | Description |
|---|---|
| GE-GN-0001 | GUBENAV – high-dynamics 100 Hz GPS and IMU sensor, on CAN |

**Package contents**: sensor, USB-C cable, instruction manual, GUBELLINI DataStudio software.

## 8.1 Related products

| Code | Description |
|---|---|
| GE-GL-0001 / 0003 | GUBELOG-01 / GUBELOG-03 – 38-channel data logger with GPS, IMU, analog and digital inputs and CAN |
| GE-ML-0001 | MICROLOG-01 – CAN logger with 16 CAN channels |
| GE-CB-0001…0005 | CAN-BOARD: accelerometer, GNSS, load cell, OPTO, type K thermocouple – values transmitted on the CAN bus |

**Information and orders**
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Technical data are subject to change without notice.